Project 1 Proposal

CSE 586: Computer Vision II

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Prediction Method Network Architecture

For this project, we can see that the primary feature to focus on is the sequence of past body poses in order to predict a future pose. In this context, we can assume that either CNNs or Transformers would be of use. Since CNNs can be trained based on time-sequence data to map the dependency of a particular pose at any point in time to the time it is executed in, it would be a good candidate. However, CNNs may not be able to capture the relations between a particular pose and another if they are separated by a large difference in time, as in a time series based dataset, there are bound to be dependencies on earlier points in time. This would require a deep CNN, which we would like to avoid for ease of training and fine-tuning (considering re-training costs). Hence, we consider Transformers to be the better fit.

Transformers are inherently better at handling large sequences due to their reliance on attention rather than convolutions, but do not preserve order. We think this problem can be solved by incorporating positional encoding. Further, the task is to implement an autoregressive forecasting model, so we can likely modify the last softmax layer of the traditional transformer to perform regression rather than classification.

We consider quantitative metrics such as MPJPE by calculating the Euclidean distance between predicted joint positions to the corresponding joint position in the pose (as ground truth), present in the dataset, as a global accuracy metric. The MPJPE measured at time intervals can give an idea on prediction accuracy decay over a specific interval. As proposed, MSE and MAE can also be experimented with during training (in the latent space). We will be using the difference in joint positions to that of the actual values in the dataset for all these metric, comparing the zero-velocity static poses to that of the constant velocity poses, to evaluate the accuracy in forecasting the poses.

Computational Platform

We aim to use Google Colab for our training purposes. This is mainly due to the features that make collaboration easier and the availability of faster GPUs with the Colab Pro plan that is made available for free to students. We will rely on the Westgate labs incase there is downtime with Colab.